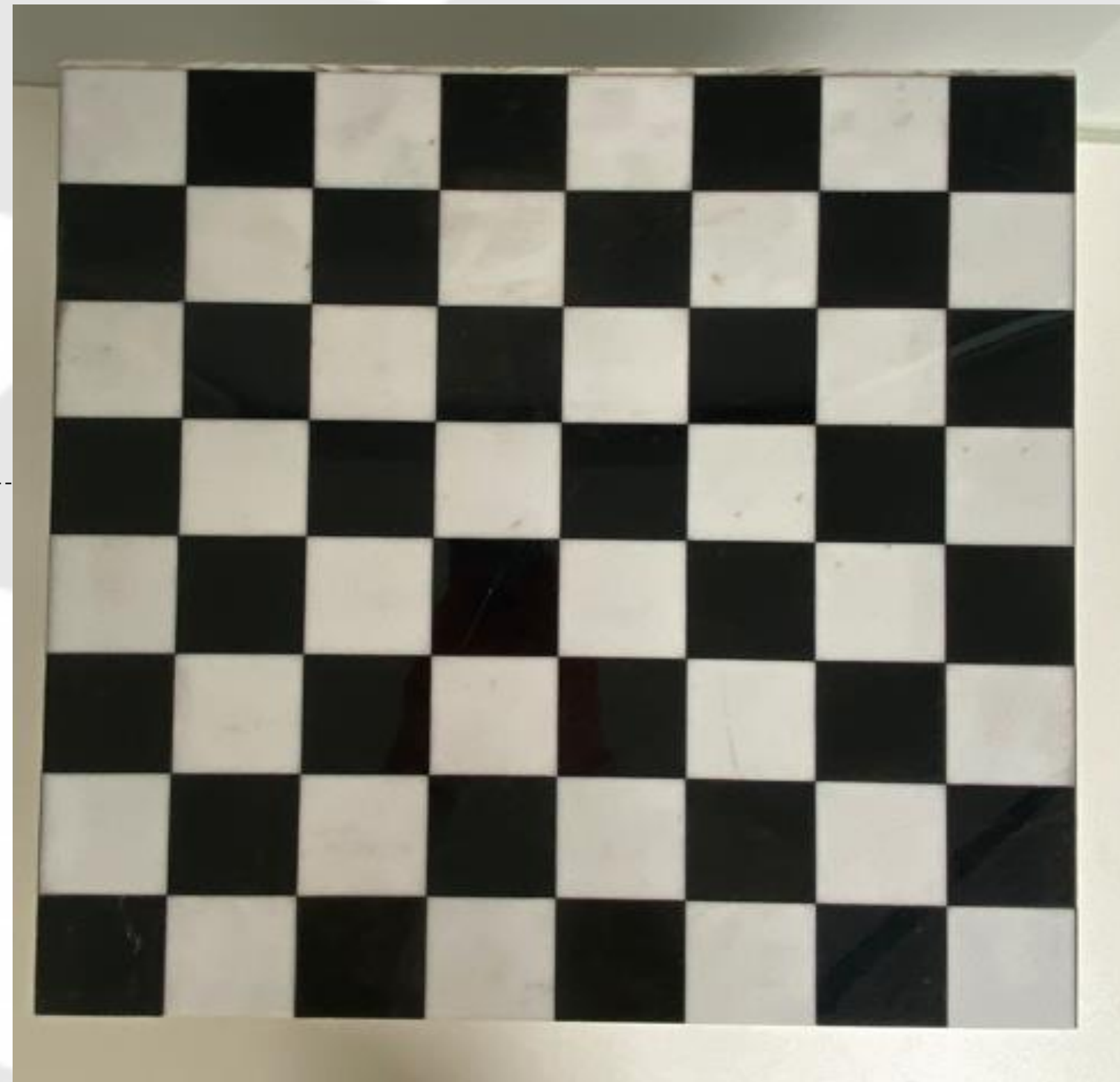


Polishing, by Hand or Machine?

Surface Reintegration Strategy of PMMA



Object Description

Artist Berend Peter Hogen

Esch (Netherlands, 1945–2021). Title Variant op een schaaks

match (Variation of chess match). Prior to 1975. PMMA, painted (or stained) wood, electric wire, metal lighting armature with a light bulb. 110x60cm. Dirt, cracks and dents

on wood, deep and shallow scratches, and noticeable abrasions on PMMA. Damage to the surface disrupts the visual effect the artist intended, so surface reintegration is necessary. The test examines different polishing techniques to smooth out abrasions.

Hand vs Machine polishing

Preparation & Methodology

Abrased PMMA



Suede microfiber cloth



Microdrill machine with soft tip

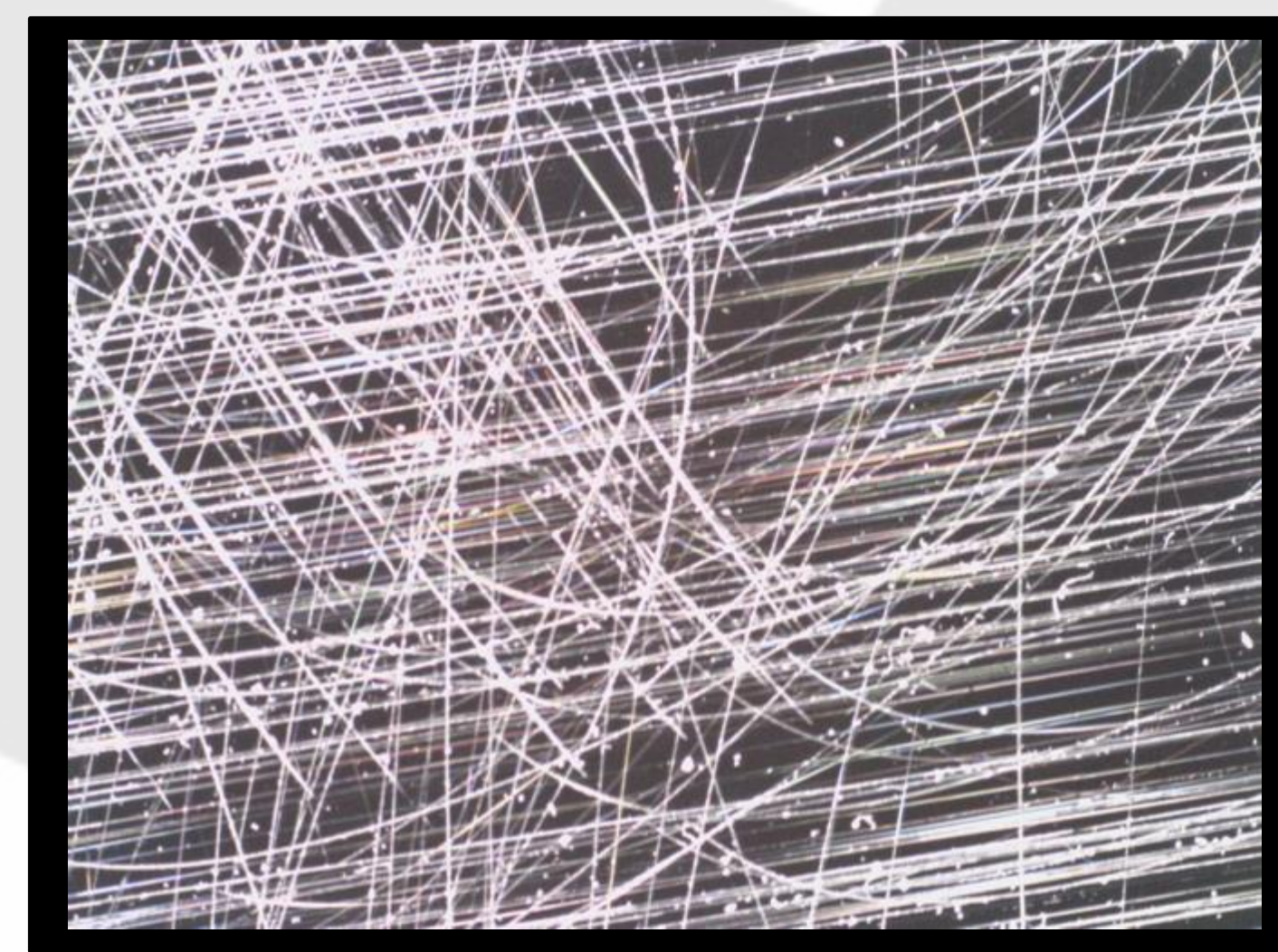


Two Polishing Pastes¹

- Novus Polish #2 Fine Scratch Remover² : contains silicon
- 3M 09639 Finesse-it Sanding Paste : silicon free

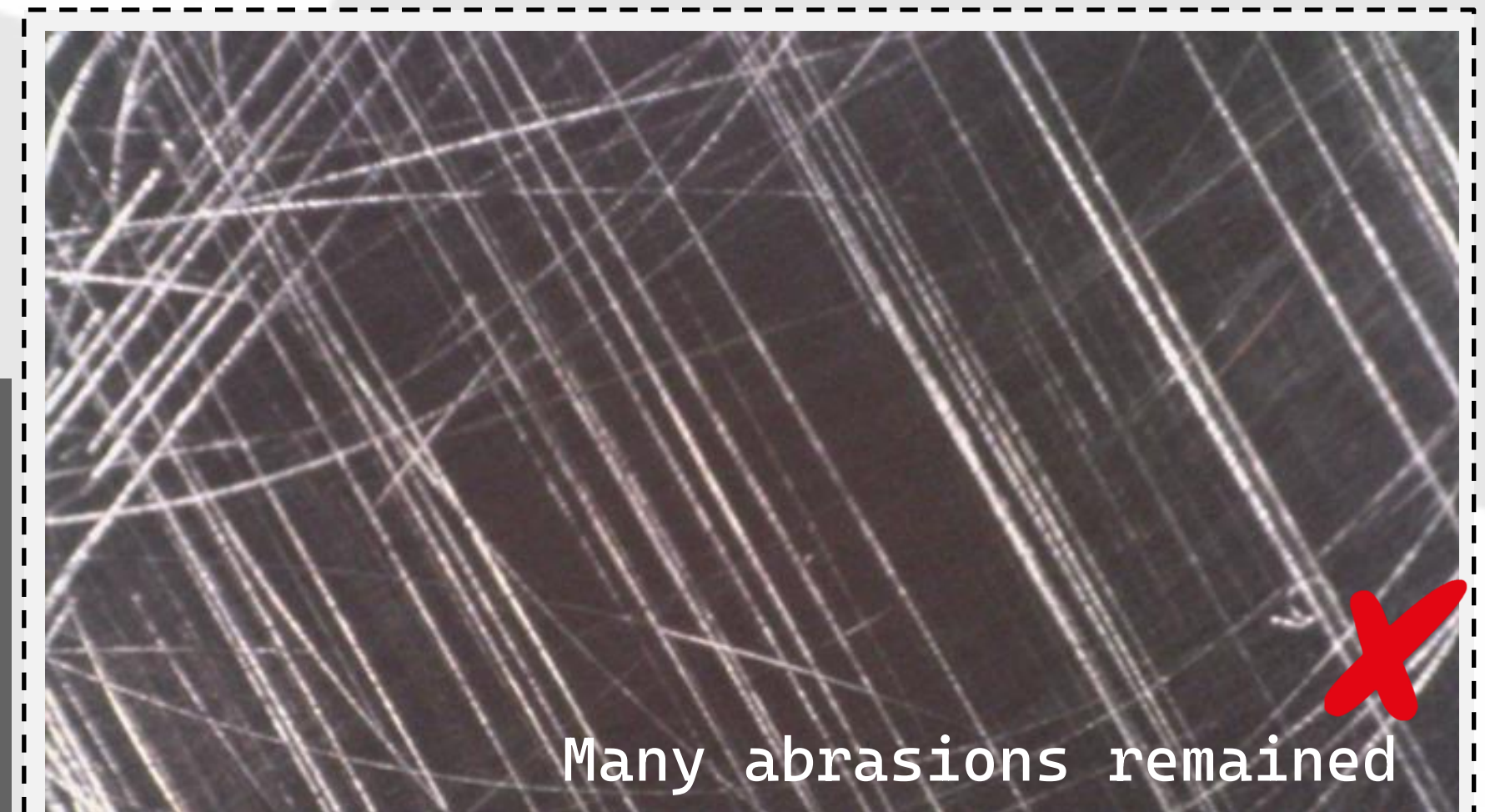
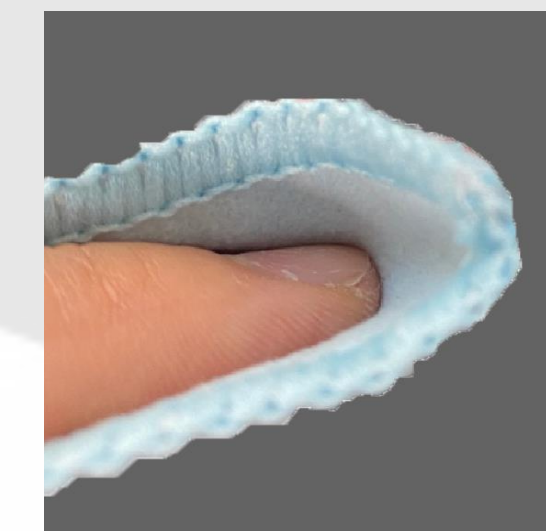
For Two minutes.

Polishing PMMA by hand and microdrill, using pastes, applying even pressure as much as possible.³



Abrased PMMA surface

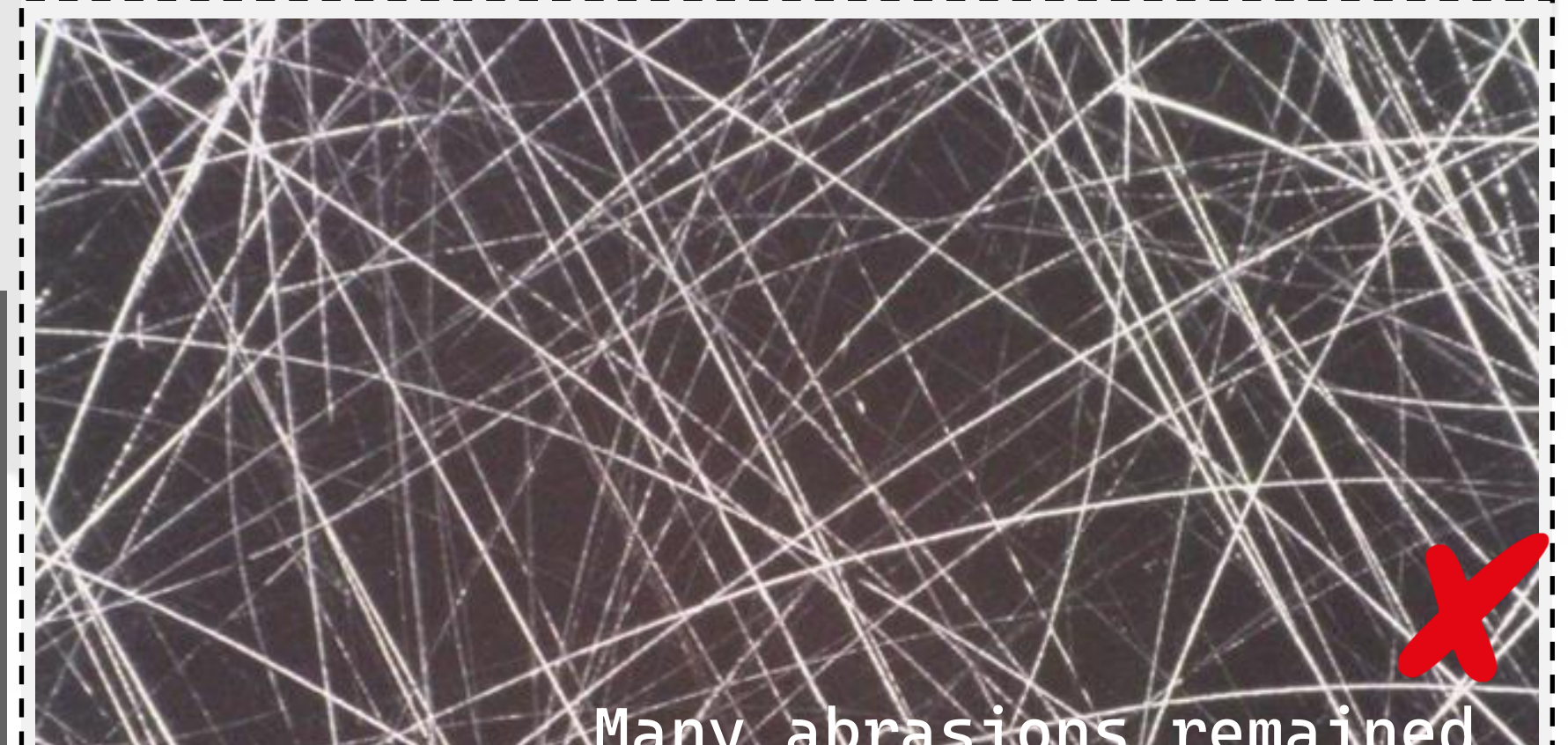
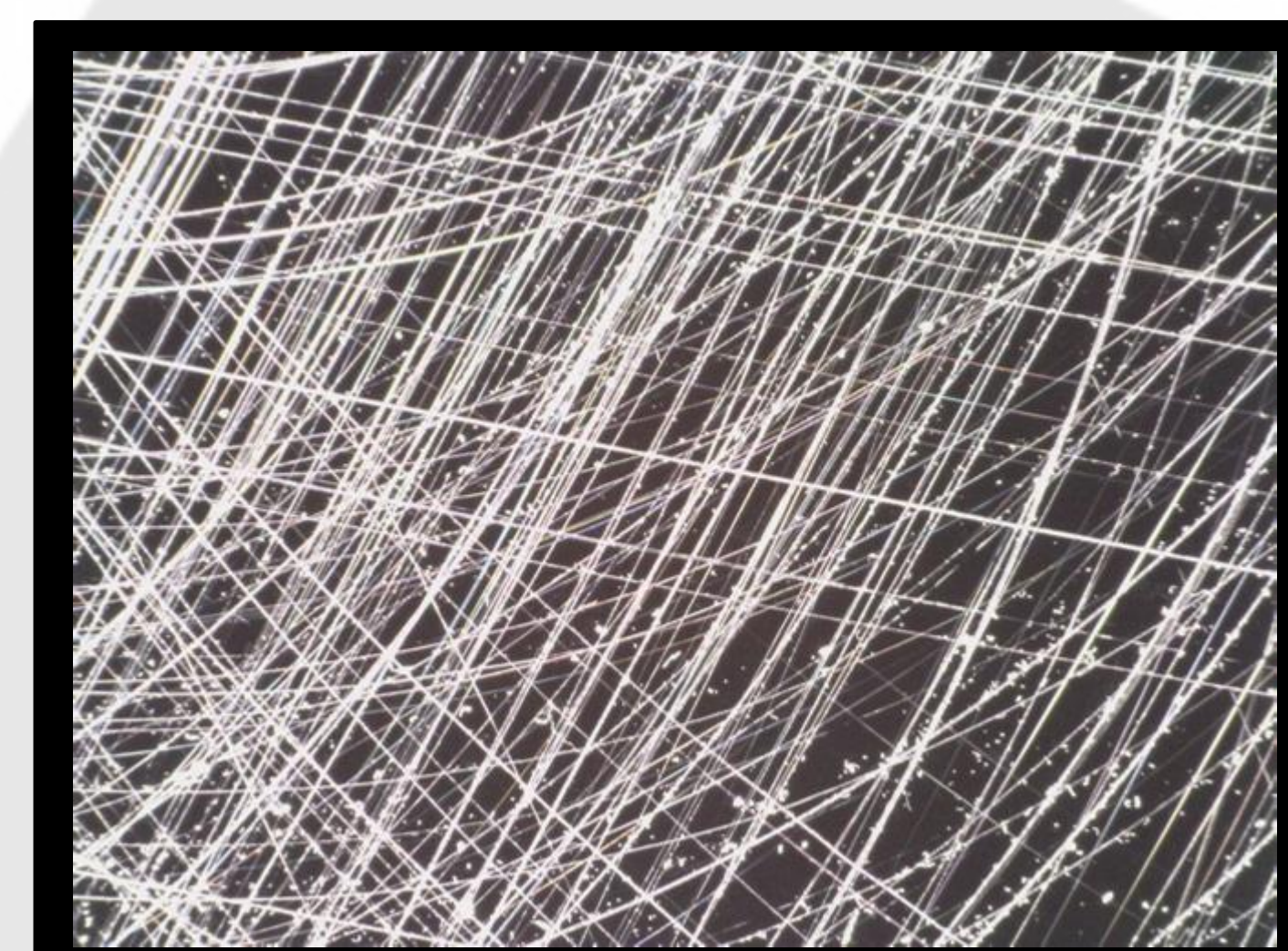
*All microscopy images in 70x magnification



Many abrasions remained



White haze, but abrasions decreased



Many abrasions remained



Good result, abrasions decreased

Checkpoints of using machine

Drill allows the user to polish PMMA fast and even. However, simultaneously, it makes micro scratches, which can cause a white haze on the surface. Besides, PMMA attracts dirt easily, so managing paste residues requires specific skills.



Scan this to see the test results with 9 different Polishing pastes!

In conclusion, polishing with a microdrill yielded a better result

than hand when taking into account the decrease in abrasions, the glossiness of the PMMA surface, efficiency in usage, and workability with pastes. One benefit of polishing by hand was that it allowed for meticulous application because the paste could only be applied to the appropriate areas, preventing debris from flying around. Nevertheless, the goal was not accomplished because the abrasive damage was not considerably reduced. On the contrary, the experiment's goal of decreasing the surface's abrasive damage was successfully accomplished when polishing with a machine. Machine polishing is advised since the gloss of the entire surface was uniformly produced, even if microabrasion was inevitable while using a machine and the surface displayed a white haze generally.

During the experiments, it became clear that it was important to identify whether the damage was scratches or abrasions before applying the appropriate techniques, such as filling or polishing.^{4,5} Based on that, a more localized approach to each method could be further explored. Additionally, studying the long-term effects of polishing paste residues on the chemical changes of the PMMA surface after mechanical polishing would be a valuable area of research in the future.⁶

Further Study

Note & Reference: 1 The two polishing pastes were selected based on: Hedwig Braam(2019). Shine on! A comparative study into polishing products for treating abraded high-gloss sculptures and an insight into ethical considerations for polishing plastics. [Research project of the University of Amsterdam/ unpublished manuscript]. 2,6 Walthew, Jessica, Sarah Barack, Nishan M Senanayake, Sarah J Lavin, Hannah G Dorris, Jennifer L.W Carter, and Ina T Martin.(2024) Imaging Analysis of Scratch Reduction in Poly (Methyl Methacrylate) (PMMA) Plastic with NOVUS Polishing Products. Studies in Conservation ahead-of-print, no. ahead-of-print: 1–14. 3 <https://www.plexiglas.de/en/service/processing/sanding-and-polishing-plexiglas> 4 Maria Cristina Lanza, Giancarlo Sidoti,Maria Elisabetta Prunas,Paola Iazurlo (2021). The decay of PMMA and its conservation treatment: The case study of a kinetic work.Transcending Boundaries: Integrated Approaches to Conservation. ICOM-CC 19th Triennial Conference, Beijing. 5 Laganà, A. N. N. A., Rivenc, R. A. C. H. E. L., Langenbacher, J. U. L. I. A., Griswold, J. O. H. N., & Learner, T. (2014). Looking through plastics: Investigating options for the treatment of scratches, abrasions, and losses in cast unsaturated polyester works of art. In ICOM-CC 17th Triennial Conference Preprints, Melbourne (pp. 15-19).

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- **Date:** Oct.2024

* The author took all the photographs unless stated otherwise.

